



5 FROM K-SPACE TO IMAGE – THE FOURIER TRANSFORM

The Fourier Transform converts *k*-space data (frequency domain) into an image (spatial domain).



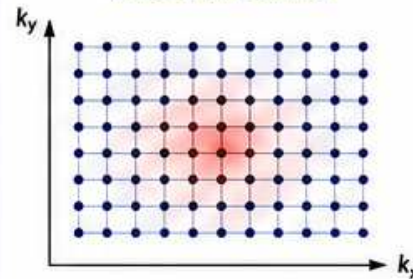
THE BIG IDEA

MRI does not measure an image directly. It measures *k*-space data. The Fourier Transform (FT) mathematically reconstructs the image you see.

THE FLOW

- 1 MRI acquires raw data in *k*-space.
- 2 The data is arranged in a k_x - k_y matrix.
- 3 A 2D Fourier Transform (FT) is applied.
- 4 The FT converts the frequency data into spatial information.
- 5 The magnitude of the result is displayed as the final image.

1. RAW K-SPACE DATA (FREQUENCY DOMAIN)



Each point (k_x , k_y) has a complex value (amplitude + phase). Center = low spatial frequencies, outside = high frequencies.

2. K-SPACE MATRIX (ARRANGED)

0.6-0.2i	-0.1+0.3i	0.2-0.1i	-0.3+0.2i	0.1+0.1i
...	0.1	...	0.5	...
...	0.7	...	0.5	...
...	-0.8+0.3i	-0.2+0.3i	-0.2i	...
...	-0.3i	-0.2+0.3i	-0.9i	...
...	-0.2+0.3i	-0.3+0.3i	-0.9i	...
...	-0.1+0.2i	-0.3i	-0.2+0.1i	...
...	...	...	...	...
...	...	...	...	...

Raw data is arranged into a 2D matrix with complex values.

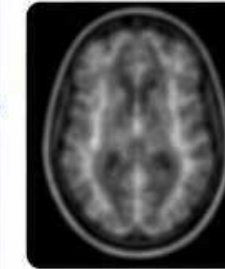
3. 2D FOURIER TRANSFORM (FT)



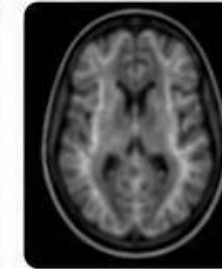
The FT converts frequency-encoded data (*k*-space) into spatial domain information.

4. COMPLEX IMAGE (SPATIAL DOMAIN)

REAL PART

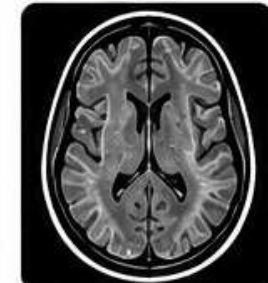


IMAGINARY PART



FT output is complex (real + imaginary). Not directly displayed.

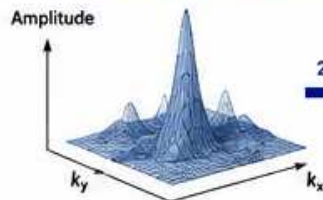
5. MAGNITUDE IMAGE (DISPLAYED)



The magnitude of the complex image is displayed. Phase information is lost.

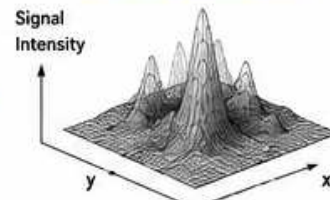
WHAT DOES THE FOURIER TRANSFORM DO?

K-SPACE (FREQUENCY DOMAIN)



2D FT

IMAGE (SPATIAL DOMAIN)



FT converts sinusoids (frequency information) into spatial locations (where structures are in the image).

FOURIER TRANSFORM – MATHEMATICAL VIEW (2D)

The 2D Discrete Fourier Transform converts *k*-space to image:

$$I(x, y) = \sum_{k_y} \sum_{k_x} S(k_x, k_y) e^{i2\pi(k_x x + k_y y)}$$

$I(x, y)$ = image intensity at position (x, y)
 $S(k_x, k_y)$ = *k*-space data (complex)
 k_x, k_y = spatial frequency coordinates
 i = imaginary unit ($\sqrt{-1}$)

In simple terms: The image is the sum of all the sine and cosine waves (*k*-space points), each with a specific amplitude and phase.

PHASE INFORMATION – WHY IT MATTERS

MAGNITUDE IMAGE
(What we see)



Used for routine diagnostic imaging.

PHASE IMAGE
(Encodes useful info)



Sensitive to field inhomogeneity, flow, susceptibility, etc.

KEY POINTS

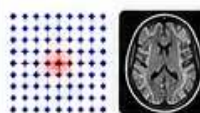
- ✓ MRI measures *k*-space, not the image.
- ✓ Center of *k*-space = overall contrast and signal.
- ✓ Outer *k*-space = detail and sharpness.
- ✓ Fourier Transform (2D FT) turns *k*-space into image.
- ✓ Magnitude is displayed; phase is usually discarded (but can be useful).

KEY PROPERTIES OF THE FOURIER TRANSFORM

- Linear**
FT of ($a + b$) = FT(a) + FT(b)
- Shift Theorem**
A shift in *k*-space → modulation in image
A shift in image → modulation in *k*-space
- Symmetry (for real images)**
Conjugate symmetry in *k*-space reduces required data.

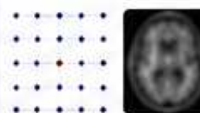
EFFECT OF K-SPACE SAMPLING ON THE IMAGE

FULL K-SPACE
(Sampled Completely)



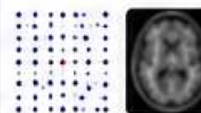
Best image quality (reference).

UNDERSAMPLING
(Less Data)



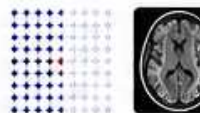
Aliasing (wrap-around), artifacts, loss of detail.

RANDOM UNDERSAMPLING
(With Acceleration)



Less aliasing, but noisier image.

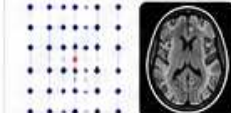
PARTIAL FOURIER
(< 100% of *k*-space)



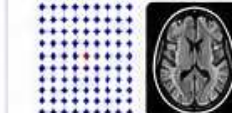
Some phase assumptions used. Faster scan.

ZERO FILLING (Interpolation in *k*-space)

BEFORE ZERO FILLING



AFTER ZERO FILLING



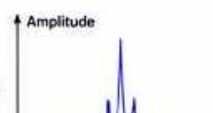
Zero filling increases matrix size (pixels) → smoother image, but DOES NOT improve true spatial resolution.

1D ANALOGY (EASIER TO UNDERSTAND)

TIME DOMAIN
(What we want)



FREQUENCY DOMAIN
(What we measure)



Just like sound: microphones measure frequencies; FT converts them back to the original sound.



BOTTOM LINE

K-space contains frequency information about the object.
 The Fourier Transform converts that information into the spatial image.
 More *k*-space data → better detail and sharpness, but takes more time.

THE MRI IMAGE RECONSTRUCTION PIPELINE



WHAT'S NEXT?

Now that we know how *k*-space becomes an image, let's explore:

- Scan time & sampling strategies (Poster #6)
- Artifacts & their *k*-space causes (Poster #7)

